

# Junlin Xiao

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## EDUCATION

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|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>University of Pennsylvania</b> , School of Engineering and Applied Science<br><i>M.S.E. in Robotics</i>                                       | Philadelphia, PA<br>Sep 2026 – Present |
| <b>University of Oxford</b> , Oxford Robotics Institute, GOALS Lab<br><i>Visiting Student in Engineering Science</i>                             | Oxford, UK<br>Sep 2024 – Jul 2025      |
| <b>The Chinese University of Hong Kong, Shenzhen</b> , School of Science and Engineering<br><i>B.Eng. in Electrical and Computer Engineering</i> | Shenzhen, China<br>Sep 2022 – Jun 2026 |

## PUBLICATIONS

### Gaussian Process Aggregation for Root-Parallel Monte Carlo Tree Search with Continuous Actions

Junlin Xiao, Victor-Alexandru Darvari, Bruno Lacerda, and Nick Hawes

Reinforcement Learning Conference (RLC), 2026

[\[arXiv\]](#) [\[PDF\]](#)

## RESEARCH EXPERIENCE

**Research Assistant**, Oxford Robotics Institute (ORI) Nov 2024 – Dec 2025  
Advisors: Prof. Nick Hawes and Dr. Victor-Alexandru Darvari Oxford, UK

- Identified limitations of voting and similarity-based aggregation in continuous-action parallel search; developed a Gaussian-process method to estimate values and select actions not sampled by the search trees.
- Led evaluations across three Gymnasium control tasks and a custom stochastic benchmark, comparing GPR2P with voting and similarity-based methods under deterministic and stochastic dynamics.
- Analyzed computational complexity and profiled module-level runtime; introduced visit-count filtering that reduced core GP regression time by approximately 100× without degrading planning performance.
- Designed ablations across algorithm components and search budgets to isolate performance gains; integrated benchmarks, algorithm interfaces, and code into ORI's internal framework for reproducibility.

**Research Intern**, Future Network of Intelligence Institute (FNii) Apr 2026 – Jul 2026  
Advisor: Prof. Shuguang Cui Shenzhen, China

- Reviewed robotic grasping methods and proposed a multimodal approach to grasp state estimation, fusing wrist and RealSense RGB-D imagery with gripper state and joint motor signals to distinguish stable grasps from failures.
- Designed a data collection protocol to align wrist and depth camera streams with joint currents and gripper state. Collected slips, unstable grasps, and false contacts under varied tabletop backgrounds and distractor objects.
- Designed a lightweight model using ResNet18 for images from multiple views and an MLP for gripper state and joint currents; used a GRU to capture changes during closure, contact, and lift to predict grasp stability.
- Built a grasping dataset by parsing UMI gripper recordings frame by frame and mapping them into Isaac. Applied 3D Gaussian Splatting to reconstruct scene appearance for Real2Sim replication.

**Research Assistant**, Shenzhen Institute of Artificial Intelligence and Robotics for Society Apr 2024 – Sep 2024  
Advisor: Prof. Jian Zhu Shenzhen, China

- Led two generations of control electronics for an anti-snoring pillow, enabling millimeter-scale air bladder adjustment based on sleep state feedback. The prototype was exhibited at the 2023 China Hi-Tech Fair.
- Designed strain gauge sensing and bridge amplifier circuitry for bladder height control; combined deformation signals with a pressure and height regression model to mitigate sensor offsets and startup drift.

## HONORS AND AWARDS

**Outstanding Graduate**, School of Science and Engineering, The Chinese University of Hong Kong, Shenzhen 2026

**China National Scholarship**, Ministry of Education of the People's Republic of China 2023 – 2024

**Dean's Honor List**, The Chinese University of Hong Kong, Shenzhen 2022 – 2024

## SKILLS

**Languages** English (fluent), Mandarin (native)

**Skills** MATLAB, C/C++, Python, ROS/ROS2, Linux, Git, Gazebo, Isaac, MATLAB, Gymnasium